Update Time: 2026.3.27

*** 代表本题未找到描述的原文本，大概翻译了个意思，题目本身是一样的**

在线预览排版可能会有些 bug，建议下载到本地。

Calculus

1. Calculate the following indefinite integral: $\int x^2 sinx\, dx$.

2. Show that the integral $\int_0^1 \frac{1}{\sqrt{sinx}} dx$ converges.

You must use the fact that the integral $\int_0^1 \frac{1}{\sqrt{x}}\, dx$ converges.

3. Calculate the following indefinite integral: $\int tan^2 x\, dx$.

4. Calculate the first derivative of the following function $\left(\sqrt[3]{x^2+x+1}\right)^e$

5. Let us suppose the function: $f(x) = e^{\frac{1}{x}}, x \neq 0$.

(1) Derive the equation of the tangent to the function at $x = -\frac{1}{2}$.

(2) Draw the graph of the function where $x < 0$.

6. Evaluate the following limit $\lim_{h \to 0} \frac{sin\, 2h^3}{h^3}$.

7. Calculate the following definite integral:$\int_e^{e^2} \frac{logx}{x}\, dx$.

8. Prove by comparing areas, that for every positive integer n, we have $\frac{1}{n+1} < log\,(1+\frac{1}{n}) < \frac{1}{n}$.

9. Calculate the following definite integral:$\int_0^{\frac{\pi}{2}} \frac{cosx}{1+\sin^2 x}\, dx$.

10*. Let us suppose the function: $f(x) = \frac{sinx}{x}$.

(1) Find the derivative of $f(x)$ and $f'(x)$.

(2) Solve $\lim_{x \to 0} f'(x)$.

11*. Find the equation of the tangent line to the curve $y = tanx$ at $x = \frac{\pi}{6}$.

12. Calculate the following indefinite integral: $\int (\log(x))^3 \, dx$.

13. Evaluate the following limit $\lim_{x \to 0} \frac{e^{4x}-1}{4x}$.

14. Calculate the following indefinite integral: $\int x \, arcsinx \, dx$.

15. Calculate the following indefinite integral: $\int cos^4 x \, dx$.

16. (1) Derive the Taylor series if $cos\,(x+h)$ to the first degree of h around x.

(2) Calculate the approximation of $cos61^\circ$ using (1). You may use $\frac{\sqrt{3}}{360}\pi \approx 0.015$. (Hint: The unit of the angle in (1) is the radian.)

17*. Draw the graph of the equation: $x - y^2 - 2y + 5 = 0$

18. Find the Taylor polynomials of third degree in $x = 0$ of $e^x + e^{-x}$.

19. Calculate the following indefinite integral: $\int \frac{x}{x+1} \, dx$.

20. Find the area under the given curves between the given bounds.

(1) $y = x^4$, between $x = -1$ and $x = 1$

(2) $y = e^x$, $x = 0$ and $x = 1$

Algebra

1. Let A be an $\mathrm{n} \times n$ square matrix, where $\mathrm{n} \geq 3$. Show the determinant of A if the (i,j)-element of A is $\mathrm{i}+\mathrm{j}$.

2. Given $\mathrm{A} = \begin{bmatrix} 1 & 2 & 3 & 5 \\ 2 & 4 & 8 & 12 \\ 3 & 6 & 7 & 13 \end{bmatrix}, \mathrm{B} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$, consider the linear system of equations: $Ax = b$.

(1) find the condition on b_1, b_2, b_3 to have a solution.

(2) Describe the nullspace of A.

3. $Q = I - 2uu^T$ is a reflection matrix when $uu^T = 1$. Two reflections give $Q^2 = I$.

(1) Show that $Qu = -u$.

(2) Find $Qv\ when\ u^T v = 0$

4. For a 3-dimentional vector $a = (1,5,3)$

(1) Calculate projection to $z - axis$.

(2) Calculate projection to $xy - plane$.

(3) Calculate projection e to the vector $b = (1,1,4)$.

5. Calculate the eigenvalues of the following symmetric matrix A. Explain the diagonalization procedure for the matrix A. $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & 0 \end{bmatrix}$

6. Find the largest number of linearly independent vectors among the six vectors below.

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} v_4 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} v_5 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} v_6 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

7. Suppose a linear T transforms $(1,1)\ to\ (2,2)$ and $(2,0)\ to\ (0,0)$. Find $T(v)$:

$$(a)\ v = (2,2) \quad (b)\ v = (3,1) \quad (c)\ v = (-1,1) \quad (d)\ v = (a,b)$$

8. Which of these transformations are not linear? The input is $v = (v_1, v_2)$:

$$(a)\ T(v) = (v_2, v_1) \quad (b)\ T(v) = (v_1, v_1) \quad (c)\ T(v) = (0, v_1)$$

$$(d)\ T(v) = (0,1) \quad (e)\ T(v) = v_1 - v_2 \quad (f)\ T(v) = v_1 v_2$$

9. If a 3 by 3 matrix has $detA = -1$, find $\det(\frac{1}{2}A)$and $\det(-A)$ and $\det(A^2)$ and $\det(A^{-1})$.

10*. Prove that if a row of a matrix A is $v = (1,0,-1)$, then v is not contained in the null space of A.

11*. Given $A = \begin{bmatrix} 0.2 & 0.4 \\ 0.6 & 0.8 \end{bmatrix}$. Find the eigenvalues and eigenvectors of the given matrix. Also, determine the matrix raised to the power of infinity.

12. Project $b = (0,8,8,20)$ onto the line through $a = (1,1,1,1)$. Find $\hat{x} = \dfrac{a^T b}{a^T a}$ and the projection $p = \hat{x}a$. Check that $e = b - p$ is perpendicular to a, and find the shortest distance $\|e\|$ from b to the line through a.

13. Let us suppose that $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, $A^2 = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$

(1) Derive the eigenvalues and eigenvectors of A.

(2) Derive the eigenvalues and eigenvectors of A^2, A^{-1}, and $A + 4I$ using the result of (1), where I is the 2×2 identity matrix.

(3) Show that the trace and determinant of A are $\lambda_1 + \lambda_2$ and $\lambda_1\lambda_2$, respectively, where λ_1, and λ_2 are the eigenvalues of A.

14. (1) If A has four orthogonal columns of lengths 1,2,3,4, what is $A^T A$

(2) If A and B are orthogonal matrices, show that their product AB is also an orthogonal matrix.

15. Find the eigenvalue of A and B and AB and BA:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \text{ and } AB = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \text{ and } BA = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

(1) Are the eigenvalues of AB equal to eigenvalues of A times eigenvalues of B?

(2) Are the eigenvalues of AB equal to the eigenvalues of BA?

16. Given matrices A and B as follows.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

(1) Calculate eigenvalues of A, B, AB and BA.

(2) Does multiplying an eigenvalue of A by an eigenvalue of B yield an eigenvalue of AB?

(3) Do AB and BA have the same eigenvalues?

17*. Consider the following augmented matrix $A = \left(\begin{array}{ccc|c} 2 & 3 & 1 & 7 \\ 2 & 5 & 3 & 17 \\ 0 & 2 & a & b \end{array}\right)$

(1) For which values of a are the columns of A linearly independent?

(2) For what condition on b does the linear system $Ax = c$ have a solution?

18. Answer the following questions:

(1) If $a_{11} = a_{22} = a_{33} = 0$, how many of the six terms in $\det A$ will be zero?

(2) If $b_{11} = b_{22} = b_{33} = b_{44} = 0$, how many of the 24 terms $b_{1j}b_{2k}b_{3l}b_{4m}$ are sure to be zero?

19. Find the line $y = C + Dx$ that minimizes the sum of the square of the distance between the line and three points $(-1,7), (1,7),$ and $(2,21)$ on the XY-plane. And draw the line.

写在最后：希望参加考试的同学如果遇到了过去问中没有的题能私信笔者，毕竟这份过去问也是依靠每个参加过考试的人才能整理出来的。